

Aditya Srivastava

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Education

Guru Gobind Singh Indraprastha University, New Delhi

2023 – 2027(Expected)

Bachelor of Technology, Electronics and Communication Engineering

CGPA: 8.75 / 10

Work Experience

Summer Trainee, DRDO

June'25 – July'25

Delhi, India

STM32 Driver Development — [GitHub Link](#)

- Developed **bare-metal SPI, I2C, and USART drivers** for the STM32F446RE, handling peripheral initialization, clock configuration, and interrupt-driven communication without HAL.
- Designed **modular and reusable drivers** with clean APIs separating initialization, data transfer, and error handling for easy integration across peripherals.
- Validated functionality through **hardware-in-the-loop testing** and debugged protocol issues including I2C ACK/NACK handling and SPI clock polarity mismatches using STM32CubeIDE.
- Technologies:** Embedded C, STM32Cube IDE, STM32 Nucleo-F446RE.

Projects

Swasthify — [GitHub Link](#) — [Live Link](#)

March'25 – April'25

- Technologies:** React, Node.js, Express, PostgreSQL, Prisma ORM, Redis, Socket.IO, Gemini AI, TailwindCSS, JWT.
- Built a multi-hospital healthcare platform with **3 role-based portals** and **31 REST API endpoints** for patients, doctors, and nurses.
- Implemented **5-vital anomaly detection** with real-time alerts using **Socket.IO** and AI-assisted clinical analysis via **Gemini API**.
- Integrated **Redis caching** to eliminate redundant AI calls and architected a cross-hospital patient record system with **JWT role guards**, refresh token rotation, and write ownership enforcement.

Amazon Go — [GitHub Link](#) — [Live Link](#)

June'26

- Technologies:** Next.js 14, React 18, TypeScript 5, Tailwind CSS, Node.js, AWS Lambda, API Gateway, DynamoDB (9 tables), Amazon Cognito, Amazon Bedrock (Claude 3 Haiku), Serverless Framework v3
- Built a **Quick-commerce platform** with **AI-first shopping**, enabling product discovery through photo, voice, and text inputs instead of traditional search.
- Deployed a scalable **AWS** architecture using Amplify, Lambda, API Gateway, and **Cognito JWT authentication** for secure user access.
- Engineered a real-time **per-pincode inventory system** with **DynamoDB Streams** across 9 tables and a 3-tier cart engine with a **Bedrock Claude 3 Haiku** adapter layer, enabling seamless switching between rule-based and LLM modes.

Technical Skills

Programming Languages: Java (primary), JavaScript, Python (basic)

Core CS: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Computer Networks, Operating Systems, SQL, Problem Solving

Frameworks & Libraries: Next.js, React.js, Node.js, Express.js, REST APIs, Socket.IO, Prisma ORM

Databases & Caching: PostgreSQL, MongoDB, DynamoDB, Redis

Cloud & Infrastructure: AWS Lambda, API Gateway, Amplify, Amazon Cognito, Serverless Framework

AI & Integrations: Amazon Bedrock (Claude 3 Haiku), Gemini AI

Developer Tools: Git, GitHub, VS Code, Postman

Achievements & Leadership

- Won **Second Prize** among **50+ teams** at Confluence 5.0, organized by BVCOE, New Delhi.
- Web Development Head**, BVP-IOSC - Led initiatives and mentored **60+ junior developers**.
- Presented research at **ICMSCI**, organized by PGDAV College (2026).